

Mobile device security and forensics unit 4 - 6

Question 1

1. When investigating mobile devices, one of the critical tasks is recovering deleted data, even when data appears to be deleted, traces of it often remain on the device, and with the right tools and techniques, this data can be recovered. List and explain Common Data Recovery Techniques [10]
2. List and explain at least 4 tools for data recovery [6]
3. Deleted Messages, Call Logs, and Media Files are one of the most commonly deleted items from mobile devices write few notes about them and explain their recovery technique and tools used if applicable [5]
4. Define Artifacts, Artifact analysis and types of Artifacts [10]
5. List and explain artifacts reconstruction [6]
6. List at least 4 tools for artifacts analysis [4]

Question 2

1. Define Mobile Network forensics [2]
2. List and explain mobile network protocols, including important evidence which can be extracted [10]
3. Mobile network forensics heavily relies on analyzing network logs and metadata to reconstruct user activities and interactions with mobile networks. List and explain at least 4 types of network logs. [4]
4. Explain the types of metadata and list at least 3 tools for metadata analysis [6]
5. Define Wi-Fi Forensics [2]
6. List and explain evidence found in Wi-Fi forensics [6]
7. List and explain tools for Wi-Fi forensics [2]
8. Define Bluetooth Forensics [4]
9. What are some of the challenges in Bluetooth forensics [5]
10. List Tools for Bluetooth Forensics [2]

Question 3

1. Define mobile device forensics [8]
2. List and explain ways of mobile applications data storage [3]
3. Mobile apps often requests access to camera, microphone, contacts, call logs, location. What is the significance of permission in forensics [4]
4. Define Reverse engineering in mobile apps [2]
5. Explain reasons for reverse engineering [3]
6. list at least 4 tools for reverse engineering [4]
7. Define malware analysis in mobile apps [2]
8. List and explain types of mobile malware [4]
9. List and explain at least 3 malware analysis process [6]
10. List any tools for malware analysis [3]

Question 4

1. Define cloud storage [2]
2. List and explain challenges in cloud forensics [12]
3. Explain platforms used in cloud forensics [3]
4. List Tools for cloud data extraction [2]
5. Explain what IOT devices is all about and what it includes [5]
6. Explain at least 5 IOT forensic challenges [10]
7. List some of the IOT Forensic evidence [5]
8. List Forensic tools for IoT [3]
9. Define Cloud-based evidence retrieval [2]
10. List challenges in cloud evidence retrieval [3]
11. List and explain cloud evidence retrieval technique [6]
12. Digital evidence must meet legal standards before being accepted in court, list and explain requirement for admissibility of evidence [5]
13. What are some of the challenges in evidence admissibility [3]
14. What are some of the legal framework governing digital evidence [3]
15. Mobile forensic investigators must comply with what law? Mention any 3 [3]
16. List and explain at least 4 ethical principles in forensics [4]
17. List any 3 common ethical challenges [3]
18. Define Incidence response procedures [2]
19. List and explain phases of incident response in order [6]
20. Mobile device security policies why is it important [2]
21. List and explain at least 5 elements of mobile security policy [10]
22. List at least 5 best practices for securing mobile devices [5]